

Corporate Finance — Illustrated Exam: Full Model Answers

With Course Formulas & TI-84 Plus Instructions

QUESTION TYPE 1 — Multiple Choice

MC Q1 — Inflation-Linked Financial Product

Question: A product pays €50 at the **beginning of each quarter** for 3 years (first payment immediately). Afterwards, payments grow at **0.5%/quarter** (inflation). Real annual interest rate = 10%. What is the price today?

Answer: d. €2,550.93

Course Formulas Used

Converting annual rate to periodic rate (compounding):

$$(1 + r_{\text{annual}}) = (1 + r_{\text{periodic}})^m$$

$$\rightarrow r_{\text{quarterly}} = (1 + r_{\text{annual}})^{(1/m)} - 1$$

Present Value of an Annuity-Due (payments at beginning of period):

$$PV_{\text{annuity-due}} = PMT \times [1 - (1 + r)^{-n}] / r \times (1 + r)$$

Present Value of a Growing Perpetuity (first payment C_1 at $t=1$):

$$PV = C_1 / (r - g)$$

Step-by-Step Solution

Step 1 — Real quarterly discount rate:

$$r_{\text{quarterly}} = (1.10)^{(1/4)} - 1 = 1.02411 - 1 = 2.411\%$$

$$\text{Shortcut: } (1.02411)^4 = 1.10, \text{ so } (1.02411)^{12} = (1.10)^3 = 1.331$$

Step 2 — Phase 1: PV of annuity-due (12 payments of €50):

$$\begin{aligned} PV_1 &= 50 \times [1 - (1.02411)^{(-12)}] / 0.02411 \times (1.02411) \\ &= 50 \times [1 - 1/1.331] / 0.02411 \times 1.02411 \\ &= 50 \times [0.2487 / 0.02411] \times 1.02411 \\ &= 50 \times 10.313 \times 1.02411 \\ &= €528.09 \end{aligned}$$

Step 3 — Phase 2: PV of growing perpetuity (starting t = 12)

The first growing payment at t=12 is $C_{12} = 50 \times 1.005 = €50.25$. The growing perpetuity formula gives PV one period before the first payment, i.e. at t = 11:

$$\begin{aligned} \text{PV at } t=11 &= C_{12} / (r - g) \\ &= (50 \times 1.005) / (0.02411 - 0.005) \\ &= 50.25 / 0.01911 \\ &= €2,629.01 \end{aligned}$$

Discount back 11 quarters to t = 0:

$$\begin{aligned} PV_2 &= 2,629.01 / (1.02411)^{11} \\ &= 2,629.01 / (1.331 / 1.02411) \\ &= 2,629.01 / 1.3001 \\ &= €2,022.84 \end{aligned}$$

Step 4 — Total price:

$$\text{Price} = PV_1 + PV_2 = 528.09 + 2,022.84 = €2,550.93 \quad \checkmark$$

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Phase 1 (annuity-due): 2nd → FINANCE → TVM Solver Set: N=12 |

I%=2.411 | PMT=50 | FV=0 | P/Y=1 Switch to BEGIN mode: 2nd → BGN →

2nd → SET Cursor to PV → ALPHA → SOLVE → **PV = 528.09**

Phase 2: Enter directly: $50 \times 1.005 \div (0.02411 - 0.005) \rightarrow$ store result
 Then $\div 1.02411^{11} \rightarrow$ **2022.84** Add: $528.09 + 2022.84 \rightarrow$ **€2,550.93**

MC Q2 — Macaulay Duration

Question: 2-year bond, YTM = 3%, coupon = 5% annually, redeemed at par. Macaulay duration?

Answer: d. 1.95 years

Course Formula Used

Macaulay Duration — weighted average time to receive the bond's cash flows (in PV terms):

$$\text{Duration} = \sum [t \times \text{PV}(\text{CF}_t)] / \text{Bond Price}$$

where:

$$\text{PV}(\text{CF}_t) = \text{CF}_t / (1 + \text{YTM})^t$$

$$\text{Bond Price} = \sum \text{PV}(\text{CF}_t) = \sum \text{CF}_t / (1 + \text{YTM})^t$$

Step-by-Step Solution

Assume par value = €100.

Bond cash flows:

$$t = 1: \text{CF}_1 = \text{coupon} = €5$$

$$t = 2: \text{CF}_2 = \text{coupon} + \text{par} = 5 + 100 = €105$$

Step 1 — PV of each cash flow:

$$\text{PV}(\text{CF}_1) = 5 / (1.03)^1 = 5 / 1.03 = €4.854$$

$$\text{PV}(\text{CF}_2) = 105 / (1.03)^2 = 105 / 1.0609 = €98.973$$

Step 2 — Bond Price:

$$\text{Bond Price} = 4.854 + 98.973 = \text{€}103.827$$

The bond trades above par (coupon 5% > YTM 3%) — makes sense.

Step 3 — Macaulay Duration:

$$\begin{aligned} \text{Duration} &= (1 \times 4.854 + 2 \times 98.973) / 103.827 \\ &= (4.854 + 197.946) / 103.827 \\ &= 202.800 / 103.827 \\ &= 1.953 \approx 1.95 \text{ years } \checkmark \end{aligned}$$

Duration < Maturity (2 years) because the bond pays coupons before maturity, pulling the weighted average time forward.

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Bond price: TVM Solver: N=2 | I%=3 | PMT=5 | FV=100 → Solve PV → **103.827**

Duration: $5 \div 1.03 \rightarrow 4.854$ $105 \div 1.03^2 \rightarrow 98.973$ $(1 \times 4.854 + 2 \times 98.973) \div 103.827 \rightarrow \mathbf{1.953 \text{ years}}$

MC Q3 — DDM — Company Z

Question: 2M shares, net profit = €3.6M, payout = 35%, ROE = 12%, cost of equity = 10%. Expected share value?

Answer: b. €31.50

Course Formulas Used

Earnings Per Share:

$$\text{EPS} = \text{Net Profit} / \text{Number of Shares}$$

Dividend growth rate (Gordon-Shapiro):

$$g = \text{ROE} \times \text{Plowback ratio}$$

$$\text{Plowback ratio} = 1 - \text{Payout ratio}$$

Constant Growth DDM (Gordon-Shapiro model):

$$P_0 = \text{Div}_1 / (r - g)$$

where Div_1 is the next dividend and r is the cost of equity.

Step-by-Step Solution

Step 1 — EPS:

$$\text{EPS} = \text{€}3,600,000 / 2,000,000 \text{ shares} = \text{€}1.80$$

Step 2 — Dividend (Div_1):

$$\text{Div}_1 = \text{Payout ratio} \times \text{EPS} = 35\% \times \text{€}1.80 = \text{€}0.63$$

Step 3 — Growth rate:

$$\text{Plowback ratio} = 1 - 0.35 = 0.65$$

$$g = \text{ROE} \times \text{Plowback} = 12\% \times 65\% = 7.8\% \approx 8\%$$

The exam rounds g to 8% — a standard simplification used in the course.

Step 4 — Share price:

$$\begin{aligned} P_0 &= \text{Div}_1 / (r - g) \\ &= 0.63 / (0.10 - 0.08) \\ &= 0.63 / 0.02 \\ &= \text{€}31.50 \quad \checkmark \end{aligned}$$

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$$0.63 \div (0.10 - 0.08) \rightarrow \text{€}31.50$$

MC Q4 — Sharpe Ratio of Portfolio

Question: Portfolio A ($w=35\%$, $E(R)=12\%$, $\sigma=15\%$) + B ($w=65\%$, $E(R)=15\%$, $\sigma=20\%$), $\rho=0.6$, $r_f=0.25\%$. Sharpe ratio?

Answer: b. 0.82

Course Formulas Used

Expected return of a portfolio:

$$E(R_p) = \sum w_i \times E(R_i) = w_A \times E(R_A) + w_B \times E(R_B)$$

Variance of a 2-asset portfolio:

$$\sigma^2_p = w_A^2 \times \sigma_A^2 + w_B^2 \times \sigma_B^2 + 2 \times w_A \times w_B \times \rho(A, B) \times \sigma_A \times \sigma_B$$

Standard deviation of portfolio:

$$\sigma_p = \sqrt{\sigma^2_p}$$

Sharpe Ratio:

$$\text{Sharpe} = [E(R_p) - r_f] / \sigma_p$$

Step-by-Step Solution

Step 1 — Expected portfolio return:

$$\begin{aligned} E(R_p) &= 0.35 \times 12\% + 0.65 \times 15\% \\ &= 4.20\% + 9.75\% \\ &= 13.95\% \end{aligned}$$

Step 2 — Portfolio variance:

$$\sigma^2_p = (0.35)^2 \times (0.15)^2 + (0.65)^2 \times (0.20)^2 + 2 \times 0.35 \times 0.65 \times 0.6 \times 0.15 \times 0.20$$

$$\text{Term 1: } 0.1225 \times 0.0225 = 0.002756$$

$$\text{Term 2: } 0.4225 \times 0.0400 = 0.016900$$

$$\text{Term 3: } 2 \times 0.35 \times 0.65 \times 0.6 \times 0.03 = 0.008190$$

$$\sigma^2 p = 0.002756 + 0.016900 + 0.008190 = 0.027846$$

Step 3 — Portfolio standard deviation:

$$\sigma p = \sqrt{0.027846} = 16.687\%$$

Step 4 — Sharpe ratio:

$$\begin{aligned} \text{Sharpe} &= (13.95\% - 0.25\%) / 16.687\% \\ &= 13.70\% / 16.687\% \\ &= 0.821 \approx 0.82 \quad \checkmark \end{aligned}$$

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$$\begin{aligned} &0.35^2 \times 0.15^2 + 0.65^2 \times 0.20^2 + 2 \times 0.35 \times 0.65 \times 0.6 \times 0.15 \times 0.20 \\ &\rightarrow 0.027846 \quad \text{2nd} \rightarrow \sqrt{} \rightarrow 0.027846 \rightarrow \mathbf{0.16687} \quad (0.1395 - 0.0025) \div 0.16687 \\ &\rightarrow \mathbf{0.821} \end{aligned}$$

MC Q5 — Portfolio Risk (Variance)

Question: Portfolio A ($w=35\%$, $E(R)=15\%$, $\sigma=20\%$) + B ($w=65\%$, $E(R)=18\%$, $\sigma=22\%$), $\rho=0.7$. Which best quantifies portfolio risk?

Answer: c. variance = 3.936%

Course Formula Used

Variance of a 2-asset portfolio:

$$\sigma^2 p = w_A^2 \times \sigma_A^2 + w_B^2 \times \sigma_B^2 + 2 \times w_A \times w_B \times \rho(A, B) \times \sigma_A \times \sigma_B$$

Step-by-Step Solution

$$\begin{aligned} \sigma^2 p &= (0.35)^2 \times (0.20)^2 + (0.65)^2 \times (0.22)^2 + 2 \times 0.35 \times 0.65 \times \\ &0.7 \times 0.20 \times 0.22 \end{aligned}$$

$$\text{Term 1: } 0.1225 \times 0.0400 = 0.004900$$

$$\text{Term 2: } 0.4225 \times 0.0484 = 0.020449$$

$$\text{Term 3: } 2 \times 0.35 \times 0.65 \times 0.7 \times 0.044 = 0.014014$$

$$\sigma^2 p = 0.004900 + 0.020449 + 0.014014 = 0.039363$$

$$\text{Expressed as percentage: } 0.039363 \times 100 = 3.936\% \quad \checkmark$$

$$\sigma p = \sqrt{0.039363} = 19.84\%$$

The exam expresses variance in percentage terms. Options a/b give wrong standard deviation values; option d gives wrong variance. Only option c is correct.

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$$0.35^2 \times 0.20^2 + 0.65^2 \times 0.22^2 + 2 \times 0.35 \times 0.65 \times 0.7 \times 0.20 \times 0.22$$

→ **0.039363 = 3.936%**

QUESTION TYPE 2 — Short Conceptual Questions

Q2.1 — Itsme Acquiring NextAuth: Financial Valuation

How should the acquisition price be determined?

Determining an acquisition price is a **financial valuation** exercise. The goal is to estimate the **intrinsic (fundamental) value** of NextAuth — what its future cash flows are worth in present value terms today. Three ingredients are needed (as established in CH0 of the course):

1. **Future Free Cash Flows (FCF):** Forecast the cash flows NextAuth will generate from operations over an explicit forecast horizon — revenues, costs, taxes, capex, and changes in working capital.
2. **Appropriate discount rate (opportunity cost of capital):** Discount at the rate that reflects NextAuth's **risk level** — its WACC, which must

be estimated using CAPM (for cost of equity) and the capital structure. This is not necessarily Itsme's WACC.

3. **Terminal (going-concern) value:** Beyond the forecast horizon, NextAuth continues generating cash. A terminal value is estimated using a growing perpetuity:

$$TV = FCF_{(n+1)} / (WACC - g_{\text{long-run}})$$

4. **Synergies:** The acquisition may unlock additional value (cost savings, new revenues). These should be estimated separately. The maximum price Itsme should pay = stand-alone value + synergy value.
5. **Minimum price:** What NextAuth's owners can get elsewhere (outside options). The final deal price will land between these bounds.

Recommended method: DCF model (discounting FCFs + terminal value) cross-checked with relative multiples (P/E, EV/EBITDA of comparable fintech/authentication companies).

Q2.2 — Present Value of Growth Opportunities (PVGO)

What is PVGO, why does it arise, how to estimate it?

What it means: PVGO is the portion of a stock's current price that comes from the market's expectation of **future profitable reinvestment**, rather than from current earnings alone. The course decomposes stock value into two parts:

$$P_0 = \underbrace{EPS / r}_{\text{Steady-state value (no growth, full payout)}} + \underbrace{PVGO}_{\text{Value of future growth opportunities}}$$

Why PVGOs arise: PVGOs arise when two conditions hold simultaneously:

- The firm **retains earnings** (plowback > 0), using: $g = ROE \times \text{plowback}$

ratio - The reinvested earnings earn a **return above the cost of equity**:

$$ROE > r$$

When $ROE > r$, each euro retained creates more than one euro of value. If $ROE = r$, retaining vs. paying out is irrelevant ($PVGO = 0$). If $ROE < r$, retaining destroys value (negative $PVGO$).

How to estimate:

$$PVGO = P_0 - EPS / r$$

where P_0 is the current market price (or the constant-growth DDM price = $Div_1/(r-g)$), EPS/r is the benchmark zero-growth value (100% payout perpetuity), and r is the cost of equity.

Q2.3 — Three Main Drivers of Bond Volatility

Bond volatility (modified duration) = the percentage change in bond price for a 1 percentage-point change in YTM:

$$\text{Volatility (\%)} = \text{Modified Duration} = \text{Macaulay Duration} / (1 + \text{YTM})$$

Three drivers (from the course — "ceteris paribus" for each):

1. Coupon Rate → inverse relationship with volatility Higher coupon → lower duration → lower volatility. A high-coupon bond returns more cash sooner (front-loaded cash flows). These early payments are less affected by discounting changes. Extreme case: a zero-coupon bond has duration = maturity (all cash at the end) and thus maximum volatility for a given maturity.

2. Time to Maturity → positive relationship with volatility Longer maturity → higher duration → higher volatility. Cash flows further in the future are discounted by $(1+r)^t$, where t is large. A change in r has a much larger effect on these distant cash flows than on near-term ones. Longer bonds therefore carry more interest-rate risk.

3. YTM Level → inverse relationship with volatility Lower YTM → higher volatility. When yields are low, future cash flows are discounted less, giving them a larger weight in the bond's price. A given change in

YTM therefore produces a larger price swing. Also, at low yield levels, rates are more likely to rise, amplifying downside price risk.

Q2.4 — Capital Market Line (CML) vs. Securities Market Line (SML)

The two equations from the course:

$$\text{CML: } E(R_p) = r_f + [(E(R_m) - r_f) / \sigma_m] \times \sigma_p$$

$$\text{SML: } E(R_i) = r_f + \beta_i \times [E(R_m) - r_f]$$

The essence of the CML: The CML formalises the positive relationship between **total risk (σ)** and expected return **for portfolios**. It represents the best possible risk-return combinations achievable by mixing the market portfolio with the risk-free asset. Any rational investor should hold a portfolio on this line — choosing a portfolio below the CML is irrational because you could earn more return for the same total risk. The slope of the CML equals the **Sharpe ratio of the market portfolio**: reward per unit of total risk.

The essence of the SML: The SML formalises the positive relationship between **systematic risk (β)** and expected return **for individual securities**. Its key insight: **only beta (market risk) is priced**, because idiosyncratic (unique) risk can be diversified away at no cost. Investors receive no compensation for carrying avoidable risk. A security plotting above the SML is underpriced (buy); below the SML is overpriced (sell or avoid).

Key distinction: CML uses total risk (σ) and applies to portfolios. SML uses systematic risk (β) and applies to individual securities.

Q2.5 — Diversification

How it works: Diversification exploits the **imperfect correlation between assets**. The course variance formula shows this directly:

$$\sigma^2 p = w_A^2 \times \sigma_A^2 + w_B^2 \times \sigma_B^2 + 2 \times w_A \times w_B \times \rho(A, B) \times \sigma_A \times \sigma_B$$

When $\rho < 1$, the cross-term is smaller than in the case $\rho = 1$ (perfect correlation), making portfolio risk **lower** than the weighted average of individual risks. The lower the correlation, the greater the risk reduction. Theoretically, if $\rho = -1$, all portfolio risk can be eliminated.

Type of risk eliminated:

- **Idiosyncratic / unique / non-systematic risk** — firm-specific factors (product recalls, management changes, lawsuits). These events are uncorrelated across firms, so they cancel out in a large portfolio. This risk **can** be eliminated through diversification.
- **Systematic / market / non-diversifiable risk** — economy-wide shocks (recessions, interest rate changes, inflation) that move all assets simultaneously. This risk **cannot** be eliminated. It is captured by beta and is the only risk for which investors are compensated (per the SML).

QUESTION TYPE 3 — Large Quantitative Exercise (SRB)

Financial data given:

Item	Value
Outstanding shares	5,000,000
Book value / share	€7
Market value / share	€8
Market value of debt	€60,000,000
Cost of debt (rd)	7%
Tax rate (t)	30%
Cov(SRB returns, market returns)	0.04
Expected market return (Rm)	9%

Item	Value
Std dev of market return (σ_m)	14%
Risk-free rate — ST gov. bond (r_f)	5%

Part a — Estimate SRB's Cost of Capital (WACC)

WACC is SRB's **opportunity cost of capital** — the minimum return the company must earn on any investment to create value for its investors. It is the correct discount rate for evaluating investment projects.

Course Formulas Used

Beta (systematic risk of SRB's equity):

$$\beta = \text{Cov}(R_i, R_m) / \text{Var}(R_m) = \text{Cov}(R_i, R_m) / \sigma^2_m$$

Cost of Equity via CAPM / Securities Market Line:

$$\text{COE} = r_e = r_f + \beta \times (E(R_m) - r_f)$$

WACC (Weighted Average Cost of Capital):

$$\text{WACC} = (E / (D+E)) \times r_e + (D / (D+E)) \times r_d \times (1 - t)$$

Always use **market values** for weights, not book values.

Step-by-Step Solution

Step 1 — Beta:

$$\begin{aligned} \beta &= \text{Cov}(\text{SRB}, \text{market}) / \sigma^2_m \\ &= 0.04 / (0.14)^2 \\ &= 0.04 / 0.0196 \\ &= 2.041 \end{aligned}$$

$\beta = 2.041$ means SRB's returns move approximately twice as much as the market — a high-risk, highly sensitive stock.

Step 2 — Cost of Equity (COE) via CAPM:

$$\begin{aligned} r_e &= r_f + \beta \times (E(R_m) - r_f) \\ &= 5\% + 2.041 \times (9\% - 5\%) \\ &= 5\% + 2.041 \times 4\% \\ &= 5\% + 8.163\% \\ &= 13.163\% \end{aligned}$$

Shareholders of SRB require a 13.16% annual return to be compensated for the systematic risk they bear. This is the return demanded per the SML.

Step 3 — Market Value Weights:

$$\begin{aligned} E &= 5,000,000 \text{ shares} \times \text{€}8 = \text{€}40,000,000 \\ D &= \text{€}60,000,000 \\ D + E &= \text{€}100,000,000 \\ w_E &= 40 / 100 = 40\% \\ w_D &= 60 / 100 = 60\% \end{aligned}$$

Step 4 — WACC:

$$\begin{aligned} \text{WACC} &= w_E \times r_e + w_D \times r_d \times (1 - t) \\ &= 0.40 \times 13.163\% + 0.60 \times 7\% \times (1 - 0.30) \\ &= 5.265\% + 0.60 \times 7\% \times 0.70 \\ &= 5.265\% + 2.940\% \\ &= 8.205\% \approx 8.21\% \end{aligned}$$

The after-tax cost of debt is $7\% \times 0.70 = 4.90\%$, because interest payments are tax-deductible, reducing the effective cost of debt to the firm.

Role of WACC in investment analysis: WACC is the hurdle rate for projects. Any project with $\text{NPV} > 0$ when discounted at 8.21% creates

financial value for SRB's investors. It reflects what all investors (shareholders and debt-holders) minimally require.

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$$0.04 \div 0.14^2 \rightarrow \beta = 2.041 \quad 5 + 2.041 \times (9 - 5) \rightarrow \text{COE} = 13.163\% \quad 0.40 \\ \times 13.163 + 0.60 \times 7 \times 0.70 \rightarrow \text{WACC} = 8.205\%$$

Part b — Which Proposal? (EAA Method)

Why EAA and not simply NPV?

SRB is a **going-concern** company, meaning it will need machinery indefinitely. The two proposals have **different lifespans** (4 vs. 6 years). Comparing NPVs directly would be unfair because the 6-year project simply lasts longer — not because it is necessarily more efficient.

The correct method is the **Equivalent Annual Annuity (EAA)**, which converts each project's NPV into a constant annual value, making both options directly comparable over the same time horizon (assuming identical replacement cycles forever).

Course Formulas Used

After-tax Operating Cash Flow (OCF):

$$\text{OCF} = (\text{Revenue} - \text{Costs} - \text{Depreciation}) \times (1 - t) + \text{Depreciation}$$

This can be rewritten to highlight the depreciation tax shield:

$$\text{OCF} = (\text{Revenue} - \text{Costs}) \times (1 - t) + \underbrace{t \times \text{Depreciation}}_{\text{Depreciation tax shield}}$$

Straight-line Depreciation:

$$\text{Depreciation} = \text{CAPEX} / \text{Useful life (years)}$$

Net Present Value:

$$\begin{aligned} \text{NPV} &= -C_0 + \sum [\text{OCF}_t / (1 + r)^t] \\ &= -C_0 + \text{OCF} \times \text{Annuity factor}(r, n) \quad [\text{when OCF is constant}] \end{aligned}$$

Annuity Factor:

$$\text{Annuity factor}(r, n) = [1 - (1 + r)^{-n}] / r$$

Equivalent Annual Annuity:

$$\text{EAA} = \text{NPV} / \text{Annuity factor}(r, n)$$

Proposal A — CAPEX €400,000, 4-year life**Depreciation (straight-line):**

$$\text{Depreciation} = 400,000 / 4 = \text{€}100,000 / \text{year}$$

Annual incremental income statement:

Revenue increase	= +€175,000
Operating cost increase	= - €40,000
Depreciation	= -€100,000
EBIT (pre-tax profit)	= +€35,000
Tax (30%)	= - €10,500
Net income	= +€24,500
+ Add back depreciation	= +€100,000
Annual after-tax OCF	= +€124,500

Verification via formula:

$$\begin{aligned} \text{OCF} &= (175,000 - 40,000 - 100,000) \times (1 - 0.30) + 100,000 \\ &= 35,000 \times 0.70 + 100,000 \\ &= 24,500 + 100,000 \\ &= \text{€}124,500 \quad \checkmark \end{aligned}$$

NPV of Proposal A:

$$\text{Annuity factor (8.21\%, 4 yr)} = [1 - (1.0821)^{-4}] / 0.0821 = 3.297$$

$$\begin{aligned}\text{NPV}_A &= -400,000 + 124,500 \times 3.297 \\ &= -400,000 + 410,487 \\ &= +€10,487\end{aligned}$$

EAA of Proposal A:

$$\text{EAA}_A = \text{NPV}_A / \text{Annuity factor} = 10,487 / 3.297 = €3,181 / \text{year}$$

Proposal B — CAPEX €500,000, 6-year life

Depreciation (straight-line):

$$\text{Depreciation} = 500,000 / 6 = €83,333 / \text{year}$$

Annual incremental income statement:

Revenue increase	= +€190,000
Operating cost increase	= - €40,000
Maintenance cost	= - €5,000
Depreciation	= - €83,333
EBIT (pre-tax profit)	= +€61,667
Tax (30%)	= - €18,500
Net income	= +€43,167
+ Add back depreciation	= +€83,333
Annual after-tax OCF	= +€126,500

Verification via formula:

$$\begin{aligned}\text{OCF} &= (190,000 - 40,000 - 5,000 - 83,333) \times (1 - 0.30) + 83,333 \\ &= 61,667 \times 0.70 + 83,333 \\ &= 43,167 + 83,333 \\ &= €126,500 \quad \checkmark\end{aligned}$$

NPV of Proposal B:

$$\text{Annuity factor (8.21\%, 6 yr)} = [1 - (1.0821)^{-6}] / 0.0821 = 4.594$$

$$\begin{aligned}\text{NPV}_B &= -500,000 + 126,500 \times 4.594 \\ &= -500,000 + 581,141 \\ &= +\text{€}81,141\end{aligned}$$

EAA of Proposal B:

$$\text{EAA}_B = \text{NPV}_B / \text{Annuity factor} = 81,141 / 4.594 = \text{€}17,661 / \text{year}$$

Decision

	NPV	EAA (annual value created)
Proposal A (4 yr)	€10,487	€3,181 / year
Proposal B (6 yr)	€81,141	€17,661 / year

→ **Proposal B is clearly preferred.** Its EAA is ~5.5× higher. Even though Proposal B requires a larger upfront investment, its stronger revenue profile and manageable depreciation produce far more after-tax cash flow per year. On a like-for-like annual basis, Proposal B creates nearly €14,500 more value per year.

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NPV_A: APPS → Finance → NPV(Enter: NPV(8.21, -400000, {124500, 124500, 124500, 124500})) → **€10,487**

EAA_A: TVM Solver: N=4 | I%=8.21 | PV=-10487 | FV=0 → Solve PMT → **€3,181**

NPV_B: NPV(8.21, -500000, {126500, 126500, 126500, 126500, 126500, 126500}) → **€81,141**

EAA_B: TVM Solver: N=6 | I%=8.21 | PV=-81141 | FV=0 → Solve PMT → **€17,661**

Part c — Impact of Higher Inventory on Financial Value

Yes — holding more inventory reduces the financial value generated by the project.

More inventory = **increase in Net Working Capital (NWC)**. Working capital is cash that is tied up in operations and temporarily unavailable for other uses. In project NPV analysis, a change in working capital appears as a cash flow:

$$\text{Cash flow from } \Delta\text{NWC} = -\Delta\text{NWC} \quad (\text{increase in NWC} = \text{cash outflow})$$

An increase in NWC at the start of the project is an **additional cash outflow at $t = 0$** (not captured in operating cash flows), which directly reduces NPV. If working capital continues to rise over the project's life, there are further outflows in subsequent years. At the project's end, NWC is typically recovered — but because of the time value of money, the PV of that future recovery is less than the initial outflow.

Conclusion: More inventory → higher NWC → additional cash outflow → **lower NPV** → less financial value created.

Part d — Impact of Declining Balance vs. Straight-Line Depreciation

Yes — switching to declining balance (degressive) depreciation would increase the financial value of the project.

The reasoning draws directly on the **depreciation tax shield** formula from the course:

$$\text{Depreciation tax shield} = t \times \text{Depreciation}$$

Why front-loading helps:

1. **Total depreciation and total tax paid are identical** over the project's life regardless of the method.
2. **But timing matters** — declining balance charges larger depreciation in early years:

3. **Early years:** larger depreciation → larger tax shield → more tax saved sooner
4. **Later years:** smaller depreciation → smaller tax shield → less saving, but already in the future
5. **Time value of money:** Because we discount all cash flows, tax savings received **earlier** have a **higher present value** than the same savings received later. Front-loading the depreciation therefore **increases the total PV of tax shields**, even though their nominal total is unchanged.

Conclusion: Declining balance depreciation → front-loaded tax deductions → higher PV of tax shields → **higher NPV** → more financial value created. This is a core principle from the course: "Front-loading depreciation = lower taxes in the beginning = higher financial value."

QUESTION TYPE 4 — Capita Selecta / ACTUA

Q4.1 — The GameStop Story (Feb 2021)

Part a — The Story (Short Position, Short Squeeze, Margin Requirements)

GameStop was a struggling brick-and-mortar video game retailer. Large hedge funds believed its stock was overvalued and took **short positions**: they borrowed GameStop shares, sold them immediately, and planned to buy them back later at a lower price — pocketing the difference.

Retail investors on Reddit's r/WallStreetBets noticed the extremely high short interest and began coordinated mass buying of GameStop shares. This drove the price sharply upward.

As the price rose, the short-sellers faced growing losses. Brokers enforcing **margin requirements** — minimum collateral that short-sellers must maintain at all times — forced the funds to either post more collateral or buy back ("cover") their short positions. This forced buying pushed the price even higher, triggering more forced buying in a self-reinforcing feedback loop known as a **short squeeze**.

GameStop's stock surged from ~\$20 to nearly \$500 within weeks, causing billions in losses for hedge funds.

Part b — EMH Implications

What EMH means: The Efficient Market Hypothesis (EMH) states that stock prices **reflect all available information** at all times, so no investor can consistently earn abnormal (risk-adjusted) returns. Three forms exist: weak (historical prices), semi-strong (all public information), and strong (all information including private).

The GameStop story challenges EMH: - The price surge was **not driven by any change in fundamental value** — GameStop was still a struggling retailer. Prices diverged massively from intrinsic value for weeks, which EMH says should not persist. - The event illustrates **market inefficiency** driven by **behavioral finance** forces: herding (retail investors collectively chasing momentum), FOMO, and social media coordination — irrational, not information-based. - However, once the squeeze ended, prices eventually collapsed back toward fundamental value — partially consistent with the long-run corrective tendency implied by semi-strong EMH.

Academic conclusion: GameStop is a real-world example of temporary market inefficiency. It supports behavioral finance and the adaptive market hypothesis over strict EMH, while showing that mispricings, though they can persist briefly, eventually tend to correct.

Q4.2 — Long-Call Option Position

Part a — What is a Long-Call and Why is the Profit/Loss Asymmetric?

A **long-call position** = **buying a call option**.

By paying a **premium** (the option price) today, the buyer acquires the **right — but not the obligation** — to purchase the underlying asset at the agreed **strike price (K)** on or before the **expiry date**.

The profit/loss profile is asymmetric because of the optionality embedded in the contract:

If stock price (S) at expiry < K (out of the money):

The buyer does NOT exercise (why buy at K when market price is lower?).

Option expires worthless.

Loss = premium paid ← this is the maximum possible loss (fixed and known upfront)

If stock price (S) at expiry > K (in the money):

The buyer exercises: buys the asset at K, worth S in the market.

Profit = $(S - K) - \text{premium paid}$ ← grows linearly and is theoretically unlimited

This creates the **asymmetric "hockey stick" shape**: flat at $(-\text{premium})$ for all $S < K$, then upward-sloping once S exceeds K. The defining feature of options: **limited downside, unlimited upside**.

Part b — What Risk Does a Long-Call Hedge? Example.

Risk hedged: A long-call hedges against the risk of a **future price increase** in an asset the company needs to **buy** (an input, a commodity, or a foreign currency).

Specific example:

An **airline company (e.g., Ryanair)** has large, predictable future purchases of jet fuel. If oil prices spike, its operating costs surge and margins collapse. By buying **call options on crude oil (or jet fuel)**:

- The airline **caps its maximum fuel cost** at: Strike Price K + premium paid — regardless of how high oil goes.
- The airline **still benefits** if fuel prices fall: it lets the option expire worthless and buys fuel at the lower market price (unlike a forward contract, which locks in the price in both directions).

This gives the airline cost certainty and protects its profit margin against the most significant input cost risk it faces — at the limited cost of the option premium.

End of model answers.